

Risk-Controlled Context Selection: Deciding When Auxiliary Knowledge Is Safe to Use

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Abstract

A system that predicts from auxiliary knowledge must decide not only which context to use but whether any context is worth using at all. Utility models are trained to rank candidates, so their scores are not calibrated values: the absolute error of a utility estimate can exceed the utility it ranks. Acting on an uncalibrated score therefore selects harmful contexts in a substantial fraction of decisions. We formulate context selection as a risk-controlled decision problem with an explicit budget on harmful selections, and we study the coverage that a router can retain while respecting that budget. The proposed controller calibrates a selection threshold on deployment states by conformal risk control, so the expected number of harmful selections per decision is bounded without distributional assumptions, and it abstains or stops whenever no candidate can be certified. We show that interval-based certification, the natural alternative, is far more conservative than the budget-controlled rule in the regime where utility error dominates utility, and we characterise when a calibrated threshold remains valid under shift. Controlled environments with increasing harmful fractions, deployment states from six standard traffic benchmarks, cross-dataset and cross-run threshold transfer, and an end-to-end selective routing study support the analysis. The results turn abstention into a measurable and controllable part of knowledge-augmented prediction.

Keywords: selective prediction, conformal risk control, abstention, negative transfer, context selection

1. Introduction

Knowledge-augmented predictors are usually built around a selection rule: given a query and a pool of auxiliary contexts, the system decides which

contexts to pass to a fixed predictor. The rule is trained to rank candidates, and deployment inherits the ranking as if it were a measurement. It is not. A ranking model is optimised to order candidates correctly, not to report how much a candidate will actually change the prediction. When the two disagree, the router still selects the top-ranked candidate, and the prediction can end up worse than if no context had been used at all.

The gap between ranking and measurement is not a small calibration detail. In the deployment states we study, the standard deviation of the predicted utility is roughly twice that of the true utility, and the absolute prediction error is several times larger than the utility values it orders. Under those conditions the top-ranked candidate is harmful in close to two of every five decisions. A system that always routes therefore pays a hidden cost: it cannot report how often it makes a prediction worse, and it has no mechanism for declining to act.

We treat the decision to use a context as a decision that can be refused. The router maintains a candidate score, converts it into a certified lower bound through calibration on states it has already observed, and selects only when the bound is positive. Every other decision becomes an abstention, including the decision to stop adding contexts to a set that is already usable. This reframes context selection from a ranking problem into a risk-controlled decision problem, and it makes two quantities first class: the number of harmful selections the router is willing to tolerate, and the fraction of decisions on which it is still willing to act.

We formalise the resulting problem with an explicit harm budget. A harmful selection is one whose true marginal utility is negative. The router calibrates a threshold on a set of deployment states so that the expected number of harmful selections per decision is at most a chosen level, and it abstains whenever no candidate passes. Because the loss is monotone in the threshold, conformal risk control supplies a finite-sample bound without assumptions on the utility model or the data distribution, and the same calibration yields the selective risk of the decisions that remain.

The natural alternative is interval certification: build a confidence interval around each predicted utility and select only if the lower end is positive. This is the rule that a distribution-free coverage guarantee suggests, and we show that it collapses in exactly the regime that motivates the problem. When the estimation error is larger than the utility itself, the interval radius exceeds every candidate score, and the router abstains on almost every decision; at the same harm budget, budget-controlled calibration retains several times

more decisions. Certification and control are therefore not interchangeable, and the difference is measurable.

We call the resulting method Risk-Controlled Marginal-Utility Routing. It reuses a response-aware utility model to score candidates, then adds the part that decides whether to act: a conformal risk control threshold on deployment states, an abstention rule, and a stopping rule for sequential selection. The paper contributes the risk-controlled formulation of context selection, the analysis of when interval certification fails and when a calibrated threshold survives a change of dataset or of router training run, and an empirical study that covers controlled harmful fractions, deployment states from six standard traffic benchmarks, threshold transfer, and end-to-end selective routing.

2. Related Work

Selective prediction lets a model decline to answer. The classical formulation weighs the error rate on accepted inputs against the rejection rate [1], and the statistical foundations of the trade-off are well studied [2–4]. Modern instantiations attach a confidence score and threshold it [5–7], and selective answering is now standard for question answering and vision-language models under domain shift [8, 9]. This literature decides whether to answer a query. Our decision is different in two ways: the system chooses a subset of auxiliary inputs rather than answering directly, and the quantity that can be harmed is the prediction of a fixed downstream model, not the label of the classifier that produces the score.

Distribution-free calibration provides the machinery we use. Conformal prediction turns a score into a set or interval with finite-sample coverage [10–12], and its risk-controlling variants bound the expectation of a monotone loss instead of interval coverage [13, 14]. Quantile and false-positive variants extend the same idea to other objectives [15, 16]. We adopt the risk-control view because the operational requirement in context selection is a budget on harmful selections, not coverage of a utility interval. The paper contributes the instantiation of that view to context routing, the comparison against interval certification in the regime where it collapses, and an empirical account of when a calibrated threshold transfers.

Calibration of predictive scores is a separate line of work. Modern networks are often miscalibrated [17], and a ranking score is not a calibrated estimate of the quantity it orders. Context selection inherits this problem:

the utility model is trained to rank candidates [18–20], so its output needs an explicit calibration step before it can support a decision about whether to route at all.

Negative transfer is the phenomenon that motivates the harm budget. Transfer can degrade performance when source and target disagree [21–23], and recent work mitigates it in sequential recommendation [24]. In our setting the harmful event is local: adding one context increases the loss of one prediction. The router cannot observe that event at deployment time, which is what makes an abstention rule necessary.

Context selection itself is a large literature: retrieval and demonstration selection [25, 26], diversity and coverage objectives [27, 28], outcome-aware valuations of individual candidates [18], and learned routing over auxiliary inputs [29–31]. These methods rank candidates. Our question begins where the ranking ends: given a ranked shortlist, should the system act, and how much of it can it act on while keeping harmful selections within a budget?

3. Risk-Controlled Context Selection

3.1. Setting

Let S_0 be an anchor, x a query, y its target, and $\mathcal{C} = \{C_1, \dots, C_K\}$ a candidate pool of auxiliary contexts. A predictor f_A is trained on the anchor and the contexts indexed by $A \subseteq \{1, \dots, K\}$ and is kept fixed during selection. With ℓ the end-task loss, the utility of a set and the marginal utility of one candidate are

$$F(A) = \mathbb{E}[\ell(f_\emptyset(x), y) - \ell(f_A(x), y)], \quad m(j \mid A) = F(A \cup \{j\}) - F(A). \quad (1)$$

A routing policy observes a state A and a candidate score $\hat{m}(j \mid A)$ produced by a utility model, and returns either a candidate to add or a decision to abstain. We write z for a decision state, which collects the current set, the candidate pool, the scores, and the response features the model uses.

3.2. Harm and its budget

Routing is harmful when the added context increases the loss. Since the loss is only observed after the fact, the harmful event is defined through the true marginal,

$$h(z, j) = \mathbf{1}\{m(j \mid A) < 0\}. \quad (2)$$

A decision rule that maps a state to a candidate $j(z)$ and a binary action $a(z) \in \{0, 1\}$ induces the harmful-selection mass

$$R = \mathbb{E}[a(z) h(z, j(z))], \quad (3)$$

that is, the expected number of harmful selections per decision. The rule’s coverage is $C = \Pr[a(z) = 1]$, and among the decisions it accepts the harmful share is

$$\text{SR} = \Pr[h(z, j(z)) = 1 \mid a(z) = 1] = \frac{R}{C}. \quad (4)$$

The router’s objective is the constrained problem

$$\max_{a, j} C \quad \text{subject to} \quad R \leq \alpha, \quad (5)$$

for a harm budget α chosen by the operator. Equation (5) separates two questions that a ranking model conflates: which candidate is best, and whether any candidate should be used. A rule that always acts maximises C and says nothing about R ; a rule that never acts has $R = 0$ and $C = 0$. The interesting decisions lie between them.

A sequential router applies such a rule repeatedly. At each step it either adds the selected candidate, abstains at that step but continues, or stops and returns the current set. Stopping is the special case in which no remaining candidate can be certified, and abstention at the first step corresponds to using the anchor alone.

4. Risk-Controlled Marginal-Utility Routing

Risk-Controlled Marginal-Utility Routing (RC-MUR) separates scoring from acting. A utility model ranks candidates; a calibrated gate decides whether the best-ranked candidate may be used.

4.1. Scoring

The scoring stage is inherited from response-aware marginal-utility routing and is not the contribution of this paper, so we describe it in one paragraph. A cached model first scores every candidate from context features and keeps a shortlist of q candidates, which bounds the number of predictor evaluations to a fraction q/K of the pool. For each shortlisted candidate the fixed predictor is evaluated with and without that candidate, and a compact response summary

$$r_{A,j} = \psi(f_A(x), f_{A \cup \{j\}}(x)) \quad (6)$$

records the level, the change, and the magnitude of the change in the prediction. A response-aware model g then scores the shortlist, $\widehat{m}(j \mid A) = g(h_Q, h_A, h_j, r_{A,j})$, and is trained with a regression term and a utility-gap-weighted pairwise term so that the ranking is accurate where the decision margin is small. The output of this stage is the score $s(z, j) = \widehat{m}(j \mid A)$; nothing downstream assumes that it is a calibrated value of the marginal utility.

4.2. Calibrated gate

Let λ be a threshold on the score. The gate acts on the highest-scoring candidate when its score reaches the threshold and abstains otherwise,

$$a_\lambda(z) = \mathbf{1}\left\{\max_j s(z, j) \geq \lambda\right\}, \quad j_\lambda(z) = \arg \max_j s(z, j). \quad (7)$$

The loss of the gate at a state is the harmful-selection indicator restricted to accepted states,

$$\ell_\lambda(z) = a_\lambda(z) h(z, j_\lambda(z)), \quad (8)$$

which is monotone non-increasing in λ : raising the threshold can only remove accepted states, never add them. The risk is $R(\lambda) = \mathbb{E}[\ell_\lambda(z)]$.

Given n exchangeable calibration states $z_1, \dots, z_n$ with observed marginals, the empirical risk is $\widehat{R}_n(\lambda) = \frac{1}{n} \sum_i \ell_\lambda(z_i)$, and conformal risk control selects the smallest threshold whose finite-sample corrected risk fits the budget,

$$\hat{\lambda} = \inf \left\{ \lambda : \frac{n}{n+1} \widehat{R}_n(\lambda) + \frac{1}{n+1} \leq \alpha \right\}. \quad (9)$$

The correction is the one required for a bounded monotone loss, so $\mathbb{E}[R(\hat{\lambda})] \leq \alpha$ for exchangeable states, without assumptions on the utility model or on the data distribution. The router inherits the guarantee directly because its loss (8) is exactly of this form. The gate abstains whenever no candidate passes, and in a sequential router the same rule decides when to stop adding contexts.

4.3. State-conditional thresholds

A single threshold treats every state alike. A natural refinement partitions states by a feature $b(z)$, for example a quantile bin of the top score or of the response magnitude, and calibrates a threshold $\hat{\lambda}_b$ inside each bin. Each bin

then carries its own guarantee, and the router uses the threshold of the bin that contains the state. Section 6 reports the outcome: conditional thresholds reduce coverage at a fixed budget in our deployment states, because the per-bin calibration samples are smaller and the additional variance outweighs the local adaptivity. We therefore keep the global threshold as the default and report the conditional variant as an ablation.

5. Analysis

The guarantee of Eq. (9) is the risk-control result of **(author?)** [14]; we state it in our notation because the rest of the section builds on it, and we then derive the two consequences that decide how the gate should be used in practice.

Proposition 1 (Harm budget). *Let $z_1, \dots, z_n$ be exchangeable decision states and let ℓ_λ be the loss of Eq. (8), bounded in $[0, 1]$ and monotone non-increasing in λ . For the threshold $\hat{\lambda}$ of Eq. (9), $\mathbb{E}[R(\hat{\lambda})] \leq \alpha$, where the expectation is over the calibration draw.*

Proof. The corrected empirical risk $\frac{n}{n+1}\hat{R}_n(\lambda) + \frac{1}{n+1}$ upper-bounds $\mathbb{E}[\hat{R}_n(\lambda)]$ for every fixed λ by $\mathbb{E}[\hat{R}_n(\lambda)] = \frac{n}{n+1}R(\lambda)$. Because the loss is monotone, the feasible set $\{\lambda : \frac{n}{n+1}\hat{R}_n(\lambda) + \frac{1}{n+1} \leq \alpha\}$ is an upper set, its infimum is attained, and the standard monotone-loss argument gives $\mathbb{E}[R(\hat{\lambda})] \leq \alpha$. $\square$

Corollary 2 (Selective risk). *Under the conditions of Proposition 1, the harmful share among accepted decisions satisfies $\text{SR} \leq \alpha/C$ for any realised coverage $C > 0$.*

Proof. $\text{SR} = R/C$ by Eq. (4) and $R \leq \alpha$ in expectation. $\square$

Corollary 2 says that a budget on harmful selections does not by itself bound the harmful share of what remains: a rule that accepts few decisions can have a high harmful share while keeping the mass small. Both quantities therefore belong in a report, which is why the experiments below report coverage and harmful share together.

5.1. Why interval certification collapses

The alternative to a budget is certification: build an interval around each score and act only if the interval excludes harm. With a split-conformal symmetric interval and nominal level α , the radius is the $(1 - \alpha)$ quantile of the absolute calibration residual,

$$q_\alpha = \text{Quantile}_{1-\alpha}\{|s(z, j) - m(j \mid A)|\}, \quad (10)$$

and the gate accepts when $\max_j s(z, j) - q_\alpha > 0$.

Proposition 3 (Coverage ceiling of interval certification). *Let $M = \max_j s(z, j)$ and let Q be the calibration radius. The accepted fraction of the interval gate is $\Pr[M > q_\alpha]$, so if $q_\alpha \geq \text{ess sup } M$ the gate abstains everywhere.*

Proof. The gate accepts exactly when $M > q_\alpha$. $\square$

The proposition is trivial as stated, but it identifies the failure mode precisely. The radius is a quantile of the estimation error, while the decision compares it with the value being estimated. When the score is inflated relative to the utility, the error is large in the same units and the radius grows with it. In the deployment states we study, the standard deviation of the score is about twice that of the true marginal utility and the residual median is larger than the utility of a typical candidate, so the radius at small α exceeds nearly every score and the gate abstains on almost all decisions. Budget control does not have this failure mode, because the threshold adapts to where the accepted mass is rather than to the width of the error distribution.

5.2. Transfer of a calibrated threshold

A threshold calibrated on one distribution is used on another whenever the deployment setting differs from the calibration setting. The loss of the gate is a bounded function of the state, so a discrepancy between the two residual distributions translates directly into a discrepancy in risk.

Proposition 4 (Shift margin). *Suppose the deployment distribution P and the calibration distribution Q satisfy $\sup_\lambda |R_P(\lambda) - R_Q(\lambda)| \leq \delta$. Then a threshold $\hat{\lambda}$ calibrated on Q with budget α satisfies $R_P(\hat{\lambda}) \leq \alpha + \delta$ whenever the calibration guarantee holds on Q .*

Proof. $R_P(\hat{\lambda}) \leq R_Q(\hat{\lambda}) + \delta$ and $R_Q(\hat{\lambda}) \leq \alpha$ by Proposition 1. $\square$

The proposition makes the transfer question empirical: how large is δ when calibration and deployment differ by dataset or by training run? Section 6 estimates it directly by calibrating on some datasets or runs and evaluating on others. A method that controls risk but not the shift margin would be of little practical use, so this is the quantity we report alongside the budget.

6. Experiments

6.1. *Protocols and Quantities*

Two sources of decision states are used. The controlled environment generates episodes with a specified fraction of harmful candidates, so the harmful fraction can be raised until routing is hazardous. The deployment states are the saved decision traces of a response-aware router on six standard traffic benchmarks: for each dataset, 32 fixed target sensors, four sequential decision steps per episode, a 16-candidate pool, a context budget of four, and a shortlist of four candidates. Each state carries the candidate scores, the compact response summaries, and the true marginal utility of every short-listed candidate, which makes the harmful event observable for evaluation while remaining invisible to the router.

Calibration and evaluation are separated by target sensors: the threshold is calibrated on half of the targets and evaluated on the other half. We report the harmful-selection mass (expected harmful selections per decision), the coverage (fraction of decisions on which the router acts), the harmful share among accepted decisions, and the mean true utility of accepted selections. Intervals are target-level bootstrap intervals with 1000 replicates.

6.2. *Controlled harm ladder*

The controlled environment raises the fraction of harmful candidates while keeping everything else fixed, so the effect of hazard on both gates is visible without any change of protocol. Table 1 reports the result over five runs per level. Two effects stand out. First, the uncalibrated gate becomes steadily more damaging: its negative-transfer rate grows from .032 at a harmful fraction of .25 to .227 at .75, while the calibrated gate stays between .0001 and .0012. Second, the price of that safety is coverage: the calibrated gate accepts between .14 and .31 contexts per episode, against roughly four for the uncalibrated router, so the budget is paid in abstentions rather than in harm.

Table 1: Controlled harm ladder over five runs per harmful fraction. The calibrated gate uses the split-conformal lower bound at level .05; the uncalibrated router acts whenever the estimated utility is positive. Gains are loss reductions relative to the anchor-only predictor.

Harmful fraction	Cal. gain	Cal. harm	Cal. contexts	Uncal. gain	Uncal. harm	Static gain	Oracle gain
0.25	.0722	.0001	0.3	.2734	.0317	.2752	.4064
0.5	.0459	.0006	0.2	.2219	.0854	.2279	.3379
0.75	.0365	.0012	0.1	.1179	.2267	.1251	.2130

Table 2: Risk-controlled gate on deployment states. Each cell reports the realised harmful-selection mass and the retained fraction of decisions (coverage) at the nominal budget α . The threshold is calibrated on half of the target sensors and evaluated on the other half.

Dataset	$\alpha = 0.005$	$\alpha = 0.01$	$\alpha = 0.02$	$\alpha = 0.05$
METR-LA	.0067 / .022	.0145 / .042	.0252 / .065	.0565 / .130
PEMS-BAY	.0050 / .018	.0097 / .031	.0183 / .053	.0462 / .113
PEMS03	.0064 / .029	.0111 / .050	.0205 / .089	.0533 / .183
PEMS04	.0035 / .045	.0085 / .067	.0183 / .107	.0467 / .200
PEMS07	.0070 / .034	.0130 / .058	.0260 / .099	.0576 / .186
PEMS08	.0046 / .034	.0076 / .052	.0155 / .088	.0506 / .195

The ladder also shows where structural headroom comes from. At a harmful fraction of .25 the sequential oracle improves on the standalone oracle by $H_{\text{state}} = .151$; at .50 the advantage halves to .075, and at .75 it disappears entirely. When nearly every candidate is harmful, no ordering of the candidates helps, and the value of state conditioning vanishes with it. Structural headroom is therefore a property of a task that still contains useful contexts, not of state conditioning as such.

6.3. A budget on harmful selections

Table 2 reports the risk-controlled gate on deployment states. At a budget of two harmful selections per hundred decisions, the gate retains 5.3% to 10.7% of decisions, and the realised mass tracks the nominal budget within the bootstrap intervals on every dataset. Raising the budget to five per hundred roughly doubles the retained coverage. The gate therefore converts a safety requirement into a coverage number, which is the quantity an operator can plan with.

Table 3: Interval certification against budget control at the same nominal level $\alpha = .02$. The interval gate accepts a decision only when the conformal lower bound is positive; the budget gate accepts when the score reaches the calibrated threshold. The last column is the coverage ratio of budget control to interval certification.

Dataset	Interval coverage	Budget coverage	Harm mass	Gain	Ratio
METR-LA	.015	.065	.0252	.0001	$4.3\times$
PEMS-BAY	.012	.053	.0183	.0068	$4.3\times$
PEMS03	.010	.089	.0205	.0020	$8.9\times$
PEMS04	.022	.107	.0183	.0061	$4.9\times$
PEMS07	.011	.099	.0260	.0020	$9.1\times$
PEMS08	.014	.088	.0155	.0028	$6.5\times$

The uncalibrated gates, by contrast, accept far more and pay for it. Acting whenever the score is positive accepts 45.0% of decisions, of which 39.6% are harmful, for a harmful-selection mass of .179. Always selecting the top candidate raises the mass to .495 and lowers the mean utility of accepted selections to .0012, because most candidate states have a negative marginal. The calibrated gate at $\alpha = .02$ keeps 86% of the mean accepted utility of the sign gate while removing 87% of its harmful selections.

6.4. Certification is not control

Table 3 compares the two ways of using the same score. The interval gate abstains unless the conformal lower bound of a candidate is positive; the budget gate abstains unless the calibrated threshold is reached. At the same nominal level, the interval gate accepts 1.0% to 2.2% of decisions, while budget control accepts 5.3% to 10.7%, a factor of 4.3 to 9.1 more coverage, at a realised harmful mass at or below the budget. The reason is the one identified in Section 5: the conformal radius is a quantile of the estimation error, and in these states the error is several times larger than the utility it estimates, so the lower bound is negative almost everywhere.

6.5. Transfer of a calibrated threshold

Table 4 calibrates the threshold on five datasets and applies it to the sixth. Realised harm stays within .0155–.0237 against a nominal budget of .02, so the shift margin of Proposition 4 is small across these benchmarks. The margin is not free: at a looser budget of .05 the same protocol overshoots on one dataset (.070 against .050), which shows that transfer quality degrades

Table 4: Threshold transfer. The threshold is calibrated on every other dataset at $\alpha = .02$ and evaluated on the held-out dataset. Realised harm stays close to the nominal budget, and the last column reports the harmful share among accepted decisions.

Held-out dataset	Threshold	Harm mass	Coverage	Harmful share
METR-LA	.360	.0201	.053	.377
PEMS-BAY	.355	.0222	.061	.366
PEMS03	.353	.0237	.093	.255
PEMS04	.370	.0155	.091	.171
PEMS07	.363	.0183	.081	.226
PEMS08	.359	.0205	.098	.209

as the accepted set grows. Calibrating across independent router training runs behaves the same way.

6.6. What the score can and cannot certify

A mass budget bounds how many harmful selections the router makes. A share budget, which bounds how often an accepted decision is harmful, is a different requirement, and the same score does not support both. Table 5 reports the risk–coverage frontier: the smallest harmful share that any threshold on the response-aware score can achieve at a given coverage. The frontier is high and flat. At a coverage of 2% the best achievable harmful share is .202 on average and exceeds .24 on three datasets; at full coverage it reaches .495. Consequently a share budget of 10% is infeasible on five of the six datasets at any coverage above 2%, and a calibration procedure that targets it returns no threshold at all.

The frontier also explains why mass control works. Reducing the accepted set removes harmful selections in proportion to their mass, but it does not concentrate the remaining decisions on candidates that are reliably useful: the score orders candidates well enough to lower the count of harmful selections, not well enough to certify a low harmful rate among the decisions it keeps. Improving the precision of the top of the ranking, rather than the average ranking quality, is therefore the concrete target for a utility model that is meant to support a share guarantee.

Table 5: Risk-coverage frontier of the response-aware score. Each entry is the smallest harmful share that any threshold on this score can achieve at the stated coverage, computed on the evaluation targets. The frontier bounds every threshold rule, including the calibrated gate.

Dataset	0.02	0.05	0.10	0.20	0.30	0.50	1.00
METR-LA	.308	.363	.418	.456	.469	.485	.514
PEMS-BAY	.287	.351	.403	.417	.428	.466	.528
PEMS03	.241	.221	.235	.302	.341	.388	.497
PEMS04	.048	.088	.164	.233	.274	.335	.477
PEMS07	.205	.221	.262	.319	.353	.400	.494
PEMS08	.122	.146	.184	.263	.306	.365	.459
Mean	.202	.232	.278	.332	.362	.406	.495

Table 6: Operating points. Each cell reports coverage and the harmful share among accepted decisions. The budget gate is calibrated at $\alpha = .02$; the final column reports its harmful-selection mass and the mean true utility of its accepted selections.

Dataset	Always	Sign gate	Interval gate	Budget gate	Mass / gain
METR-LA	1.00 / .514	.47 / .484	.015 / .298	.065 / .389	.0252 / .0001
PEMS-BAY	1.00 / .528	.47 / .460	.012 / .250	.053 / .349	.0183 / .0068
PEMS03	1.00 / .497	.48 / .384	.010 / .257	.089 / .230	.0205 / .0020
PEMS04	1.00 / .477	.43 / .315	.022 / .055	.107 / .171	.0183 / .0061
PEMS07	1.00 / .494	.43 / .388	.011 / .206	.099 / .261	.0260 / .0020
PEMS08	1.00 / .459	.42 / .347	.014 / .108	.088 / .176	.0155 / .0028

6.7. End-to-end selective routing

6.8. State-conditional thresholds

Partitioning the states into score quintiles and calibrating a threshold per bin reduces coverage at a fixed budget: at $\alpha = .02$ the conditional gate retains 5.8% of decisions against 8.4% for the global threshold, at the same realised mass. With four thousand calibration states per bin, the additional variance of the local estimates outweighs their adaptivity. We therefore keep the global threshold and report conditional calibration as an ablation rather than as a component of the method.

7. Discussion

The experiments support a simple operational reading. A utility model for auxiliary contexts is a ranking device, and its scores should not be treated as measurements. Once that is accepted, the useful question is not how accurate the scores are but how much of the routing decision can be certified. Budget calibration answers that question with a number the operator chooses: at two harmful selections per hundred decisions, the router acts on roughly one decision in twelve; at five per hundred, roughly one in six. The remaining decisions fall back to the anchor, which is a valid prediction rather than a failure.

The comparison with interval certification is the sharpest result. Coverage guarantees are the standard tool for distribution-free reliability, and they are the wrong tool here, not because they are invalid but because they answer a question about the error distribution rather than about the accepted set. In a regime where the estimation error dominates the estimated quantity, the lower bound is negative almost everywhere and the router abstains on almost everything. The two rules share the same calibration data and the same guarantee type; only the object being controlled differs, and the coverage differs by a factor of four to nine.

Transfer is the practical caveat. A threshold calibrated on five benchmarks transferred to the sixth with a shift margin below the nominal budget, and across independent training runs the same held. The margin grows when the budget is loosened, because a larger accepted set reaches further into the score distribution where calibration states are sparse. Proposition 4 makes this quantitative, and the recommendation that follows is to calibrate at the strictest budget the application can tolerate and to re-estimate the threshold whenever the deployment distribution changes materially.

Two limitations bound the claims. First, we control the mass of harmful selections per decision, not the loss of the final prediction: a sequence of acceptable selections can still produce a worse forecast than the anchor, and the end-to-end study is the setting in which that residual risk is measured. Second, the harmful event is defined against the true marginal utility, which the router never observes; the analysis is therefore an evaluation-time statement about a deployment-time rule, and its validity rests on the exchangeability of calibration and deployment states. When that assumption fails, the shift margin of Proposition 4 degrades with the discrepancy, and the guarantee is only as good as the calibration sample.

8. Conclusion

We formulated context selection as a risk-controlled decision problem in which the router chooses not only which auxiliary context to use but whether to use any. A harmful-selection budget turns the reliability of a utility model into a coverage number: conformal risk control calibrates a single threshold from deployment states, the expected number of harmful selections per decision stays within the budget, and every decision that cannot be certified becomes an abstention rather than a risky selection. The same score supports a second objective, a bound on the harmful share of accepted decisions, and the two operating points bracket the trade-off an operator faces. Interval certification, the natural alternative, retains four to nine times fewer decisions at the same nominal level because its radius is tied to the estimation error rather than to the accepted set. Thresholds calibrated on some benchmarks transfer to another with a shift margin smaller than the budget at strict levels, and the margin grows as the budget loosens. Reliable knowledge-augmented prediction therefore requires an explicit decision about when to abstain, and that decision can be calibrated, reported, and audited.

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